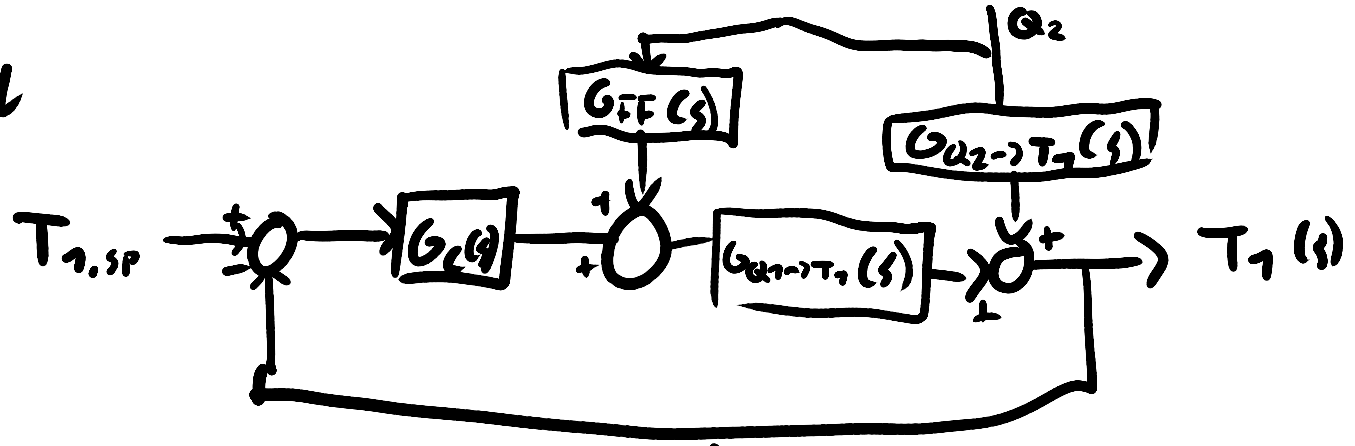
QC6: Feed Forward

Block diagram:



$$G_{FF} = \frac{G_{Q_2 \to T_1}}{G_{Q_1 \to T_1}} = -\frac{K_D}{K_P} \cdot \frac{\frac{1}{\tau_D s + 1}}{\frac{1}{\tau_I s + 1}} \cdot \frac{e^{-\theta_D s}}{e^{-\theta_P s}}$$

$\tau_D = 224\,s$
$\tau_I = 159\,s$
} Relatively close, so we'll ignore dynamics.

$\theta_D = 56\,s$
$\theta_P = 6$
} Time difference for controller to start working: 50s

$$-\frac{K_D}{K_P} = -\frac{0.744}{0.664} \approx 0.217$$